

DAC & ADC problems

Design a weighted resistor 8-bit DAC for FSV (full scale value) of 2.55V

For a ^{8-bit} DAC

$$V_o = -\frac{R_f}{R} V_{ref} [b_1 2^{-1} + b_2 2^{-2} + b_3 2^{-3} + b_4 2^{-4} + \dots + b_8 2^{-8}]$$

$$\text{Let } R_f = R = 1k\Omega$$

$$\text{Given } V_{FS} = 2.55V \quad (\text{ie}) V_o = 2.55V \text{ when } b_1 \dots b_8 \text{ is } 11111111$$

$$2.55 = -V_{ref} \left[\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \frac{1}{128} + \frac{1}{256} \right]$$
$$= -V_{ref} [0.996]$$

$$V_{ref} = -2.56V$$

To check

$$V_{FSR} \Rightarrow V_{\text{full scale range}} = -k V_{ref} = -\frac{R_f}{R} V_{ref} = 2.56V$$

$$\text{Resolution} = \frac{V_{FSR}}{2^n} = \frac{2.56}{256} = 0.01V$$

$$V_{FSV} = 2.56 - 0.01 = 2.55V$$

↓
V_{FS}

2) The basic step of a 9-bit DAC is 10.3mV. If 100000000 represents 0V. What o/p is produced if i/p is 10110111?

basic step $\Rightarrow$ resolution $\Rightarrow$ 1LSB $\Rightarrow$ 10.3mV

$$1\text{LSB} = \text{resolution} = \frac{V_{\text{FSR}}}{2^n} = 10.3\text{mV}$$

$$\therefore V_{\text{FSR}} = 10.3 \times 10^{-3} \times 2^9 = 5.27\text{V}$$

$$V_{\text{FSR}} = K V_{\text{ref}} \quad \therefore V_{\text{ref}} = 5.27\text{V} \text{ if } K=1$$

$$V_o = -5.27 \left(1 \times 2^{-1} + 1 \times 2^{-3} + 1 \times 2^{-4} + 1 \times 2^{-6} + 1 \times 2^{-7} + 1 \times 2^{-8} + 1 \times 2^{-9} \right)$$

$$\approx -3.78\text{V}$$

3) Determine the quantization error in each of the following meters

- 3 $\frac{1}{2}$ digit 200mV Full scale DVM
- 4 $\frac{1}{2}$ digit 2A full scale digital ammeter
- 6 $\frac{1}{2}$ digit 2k Ω full scale " ohmmeter

Solution :-

Resolution = 1LSB & Quantization error = $\pm \frac{1}{2}$ LSB

3 $\frac{1}{2}$ digit $\Rightarrow$ 3 full digits which can vary from 000 to 999 giving 1000 counts
 $\frac{1}{2}$ digit $\Rightarrow$ MSB can be either '0' or '1'
 $\therefore$ 0000 to 1999 giving a total of 2000 counts

Ansly for $4\frac{1}{2}$ digit 00000 to 19999 $\Rightarrow$ 20000 Counts

$6\frac{1}{2}$ digit 0000000 to 1999999 $\Rightarrow$ 2000000 Counts

a) Resolution = $\frac{V_{FSR}}{2^n} = \frac{200mV}{2000} = 1 \times 10^{-4} = 0.1mV$

Q.E = $\pm \frac{1}{2}$ LSB = 0.05mV

b) Resolution = $\frac{2}{20000} = 1 \times 10^{-4} = 0.1mA$

Q.E = $\frac{1 \times 10^{-4}}{2} = 0.05mA$

c) Resolution = $\frac{2000}{2000000} = 1 \times 10^{-3} = 1m\Omega$

Q.E = $\frac{1m\Omega}{2} = 0.5m\Omega$

4) If a 4 bit SAR type ADC ^{MSB} is stuck at zero
If FS is 15V. determine o/p from ADC for an
i/p of 5V.

Step i:- 0000 $\rightarrow$ MSB stuck at 0.
b₁ b₂ b₃ b₄

$V_{in} = 5V$ $V_{DAC} = 0$

$V_{in} > V_{DAC}$ SAR: 0000

Step ii:- Trial code 0100 $V_{in} = 5V$

$V_{DAC} = 15 \left(1 \times 2^{-2} \right) = 3.75V$

$V_{in} > V_{DAC}$ b₁ retained

SAR: 0100

Step iii:-

Total code : 0110

$$V_{in} = 5V \quad V_{DAC} = 15\left(\frac{1}{4} + \frac{1}{8}\right) = 5.625V$$

$$V_{in} < V_{DAC} \quad SAR : 0100$$

Step iv:

Total code : 0101

$$V_{in} = 5V \quad V_{DAC} = 15\left(\frac{1}{4} + \frac{1}{16}\right) = 4.6875V$$

$$V_{in} > V_{DAC}$$

Bit Retained SAR : 0101

$$O/P = 0101$$

- 5) A dual slope ADC uses a 16 bit Counter $\leftarrow 4 \text{ MHz}$
clock rate. Max i/p voltage 10V. Max. integrator o/p
voltage should be $-8V$ when the counter has cycled
through 2^N counts. The capacitor used is $0.1 \mu F$
determine the value of R of integrator. If V_{in} is
 $4.128V$ & $V_{ref} = 8V$. Find the digital o/p of
the above ADC.

Solution:

$T_i = \text{Pre-determined time period}$
 $\Rightarrow \text{Integration time}$

$$T_i = \frac{2^N \text{ Counts}}{\text{Clock rate}} = \frac{2^{16}}{4 \text{ MHz}} = 16.38 \mu s$$

$$V_{oi} \Big|_{t_i = T_i} = -\frac{1}{RC} V_{in} T_i = -8V$$

from which $RC = \frac{10}{8} \times 16.38 \times 10^{-3} = 20.47 \mu s$

$$R = \frac{20.47 \times 10^{-3}}{0.1 \times 10^{-6}} = 205 k\Omega$$

b) $V_{in} = 4.129V$ & $V_{ref} = 8V$ ~~205 kΩ~~ $x = ?$

$$V_{in} T_1 = V_{ref} T_2$$

$$V_{in} \cdot 2^N \cdot T_{clk} = V_{ref} x \cdot T_{clk}$$

$$x = \frac{V_{in} \cdot 2^N}{V_{ref}} = \frac{4.129}{8} \times 2^{16} = 33825$$

eqvt binary = 1000010000100001

b) The following values are given for a digital ramp ADC
Clock frequency 100 kHz, D/A Converter's $V_{ref} = 10.24V$
It has a 10 bit i/p.

Q determine a) digital equivalent for $V_{in} = 3.728V$,
b) Resolution of ADC, c) Conversion time required

Solution DAC $\Rightarrow$ 10 bit $V_{ref} = V_{FSR} = 10.24V$

$$\text{Step size of DAC} = \frac{10.24}{2^{10}} = 10 mV.$$

In a ramp type ADC (Counter type)

The Counter counts from 0.

Since $V_{in} = 3.728V$

1 step size is 10 mV

$$\text{Steps Required} = \frac{3.728}{10 \text{ mV}} = 372.8 \text{ (or) } 373 \text{ steps}$$

$$373 \Rightarrow 0101110101 \Rightarrow \text{digital equivalent of } 3.728 \text{ V}$$

b) Resolution : - $\frac{V_{FSR}}{2^{10}} = 10 \text{ mV}$

$$\text{In } \% \Rightarrow \frac{1}{2^{10}} = 0.1\%$$

c) 373 steps are required

373 clock pulses are required

$$1 \text{ Clock pulse} = \frac{1}{100 \text{ kHz}} = 10 \text{ } \mu\text{s}$$

$$\therefore 3730 \text{ } \mu\text{s} \Rightarrow \text{Result} \Rightarrow \text{Conversion time} = 3.73 \text{ ms}$$

7) In a system consisting of ADC & signal conditioning circuits. Total error allowed is 0.15%. If a 12-bit converter is used with following errors. Obtain the max. error which the signal conditioning circuits can have. Assume offset & gain errors are calibrated.

a) INL = 1 LSB, b) offset error = 3 LSB, c) gain error = 4 LSB

d) voltage reference error = 0.05%, e) drift error = 0.025%

Solution a) $\text{INL} = 1 \text{ LSB} = \frac{1}{2^{12}} = 0.0244\%$

b) offset error = 3 LSB = 0.0732%

c) gain error = 4 LSB = 0.0977%

voltage reference error = 0.05%

& drift error = 0.025%. Total error = 0.27%

Since offset & gain errors are calibrated total error reduces to 0.1%. $\therefore$ error allowed for signal conditioning circuits = 0.05%.